Tanmay Bhuskute

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SUMMARY

Backend Software Engineer with production experience building and operating distributed cloud infrastructure, orchestration pipelines, and AI-driven automation systems using Python, TypeScript, AWS, and Kubernetes. Drawn to early-stage startups for the chance to own systems end-to-end and ship fast; currently building agentic execution pipelines on non-deterministic LLMs, directly aligned with reliable AI orchestration at scale.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, JavaScript, TypeScript, C/C++, SQL, Go.

Web & Backend: Node.js, Express.js, Django, Flask, FastAPI, REST APIs, Microservices, WebSockets.

Databases: PostgreSQL, MySQL, DynamoDB, MongoDB, Redis.

Developer Tools: Git, GitLab CI/CD, Jenkins, AWS, Docker, Kubernetes, ArgoCD, Helm, Terraform, Linux, Bash, SonarQube.

AI & Data: Gemini, Vertex AI, LangChain, PyTorch, scikit-learn, pandas, NumPy.

WORK EXPERIENCE

Graduate Researcher | University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation, enabling deterministic closed-loop execution on top of non-deterministic VLM outputs across 20+ Android apps.
- Built device automation harness with stall detection and history-windowed decision loop, achieving **25.7x latency reduction** (39.4s vs 1012.1s avg per run) over ViBR baseline (GPT-4o + GroundingDINO + CLIP) while preserving task fidelity.

Software Development Engineer | Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, GitLab CI/CD, and HashiCorp Vault; designed Ansible based Blue Green deployment framework cutting downtime **50%** with automated rollback.
- Built self-service developer platform (dashboard + CLI/YAML automation) for environment lifecycle management, adopted by **15 engineers**, cutting provisioning turnaround **70%** and start/stop time **90%**.
- Tracked environment configs and automation job status across 40+ environments using DynamoDB; diagnosed live production issues across Linux, EC2, and containerized services via CloudWatch and Datadog correlation.

Software Developer Intern | Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

PROJECTS AND RESEARCH

PolicyPilot – Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Architected evidence-grounded **RAG pipeline** over Google Gemini 2.5-flash with citation-backed, versioned answers, and a **17-tool MCP server** exposing structured LLM tool-calling consumable by external AI agents for coverage and eligibility workflows.
- Built full-stack TypeScript backend (Express 5 REST API + MCP server) with Zod schema validation at every boundary, eliminating a class of runtime type-mismatch bugs across agent tool calls.

NYC Mobility Forecasting [PySpark, Delta Lake, scikit-learn, FastAPI] Jul 2026

- Built ML system forecasting NYC taxi/rideshare demand across **263 zones** (**2.3M-row** hourly dataset); served predictions via FastAPI backend with PostgreSQL and Redis caching, containerized with Docker.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).